

Abstract

Topic Name :- *Energy-Efficient Blockchain Consensus Algorithm*

Problem Statement :-

Conventional blockchain networks primarily use **Proof of Work (PoW) consensus mechanisms** that require extensive computational power and consume a large amount of electrical energy. This leads to high operational costs, increased carbon emissions, reliance on specialized mining hardware, and limited scalability. Such energy inefficiency presents a major obstacle to the sustainable and widespread adoption of blockchain technology.

Proposed Solution :-

To overcome these limitations, this project introduces an **Energy-Efficient Blockchain Consensus Algorithm** that replaces energy-intensive mining with a **validator-based block generation approach**. Validators are selected based on predefined efficiency, trust, and participation parameters rather than raw computational power. The algorithm utilizes lightweight cryptographic verification to achieve consensus, thereby minimizing redundant computations while maintaining essential blockchain properties such as security, decentralization, and reliability.

Outcome :-

The proposed consensus mechanism significantly reduces energy consumption, improves transaction confirmation time, and lowers hardware requirements. It provides a scalable and environmentally sustainable alternative to traditional blockchain systems, making it suitable for modern and future blockchain applications.

Conclusion :-

The proposed Energy-Efficient Blockchain Consensus Algorithm provides a sustainable alternative to traditional energy-intensive consensus mechanisms. By replacing mining with a validator-based approach, it reduces energy consumption, improves transaction speed, and enhances scalability. Despite challenges related to decentralization and validator trust, the algorithm supports the development of secure and environmentally friendly blockchain systems.

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